ACADEMY OF TECHNOLOGY

Internal Examination, Odd Sem., 2025

Paper name: Analog and Digital Electronics **Paper Code: ESC-301**
Discipline: CSE **Semester: 3rd**
Time: 1 Hour **Full Marks: 25**

Group A

1. Attempt any five questions from the following. **[1 × 5 = 5]**

i) What is the next number in octal counting sequence: 724,725,726,727?
a) 728 b) 72A c) 820 d) 730

ii) A code used for labelling the cell of K-map is
a) Natural BCD code b) Gray code c) Hexadecimal code d) octal code

iii) A decoder with an enable input can be used as
a) encoder b) parity generator c) multiplexer d) demultiplexer

iv) The maximum positive number that can be represented in 1's complement representation is
a) $2^{n-1}-1$ b) $1-2^{n-1}$ c) $-(2^{n-1}-1)$ d) 2^{n-1}

v) If $(212)_x = (23)_{10}$, where x is base (+ve integer) then the value of x is
a) 2 b) 3 c) 4 d) 5

vi) The minimum number of NAND gates required to implement the Boolean function $A + A\bar{B} + A\bar{B}C$ is equal to
a) 0 b) 4 c) 1 d) 7

vii) If xy=0, then $x \oplus y$ is equal to
a) $x + y$ b) $x' + y'$ c) xy d) $x'y'$

Group B (Attempt any four questions from the following.) [5 × 4 = 20]

2. Implement a Boolean function $F(A,B,C,D) = A\bar{B} + BD + \bar{B}C\bar{D}$ using a combination of 4:1 & 2:1 MUXs. Consider D as i/p and A, B, and C as the selected i/p to the MUX. [5]
3. Minimize the following expression in SOP form using K-map and realize the simplified function using two input NAND gates only: F(A,B,C,D)= $\sum(1,2,3,8,9,10,11,14) + d\sum(7, 15)$. [5]
4. Implement a full subtractor using OR gate & half subtractor. [5]
5. Obtain (a) Minimal SOP (b) Minimal POS for $Y = \sum_m(0,2,3,6,7) + \sum_d(8,10,11,15)$ using K-map. [3+2]
6. Design a combinational circuit using all discrete logic gates to convert Excess-3 to BCD code. [5]
7. Design a full adder using 4:1 MUX. Take carry input of adder as the input of the MUX. [5]

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Internal Examination, Odd Sem, 2025

Paper name: Data Structure and Algorithms Paper Code: PCC-CS301
Discipline: CSE Semester: 3rd
Time: 1 Hour Full Marks: 25

Group A

1. Attempt any five questions from the following. [1 × 5 = 5]

i. Consider the following operation performed on a stack of size 5.
Push(1);Pop();Push(2);Push(3);Pop();Push(4);Pop();Pop();Push(5);
After the completion of all operation, the no of element present on stack are

a) 1 b) 2 c) 3 d) 4

ii. Consider a normal queue of capacity 8 and initial value of front and rear is -1. Find the final value of rear and front after five successive insert and two delete operations.

a) front=2 and rear=5 b) front=1 and rear=4
c) front=-1 and rear=4 d) front=1 and rear=5

iii. In the Tower of Hanoi problem, if there are n disks, how many total moves are required to solve the problem optimally?

a) n b) $2n$ c) $2^n - 1$ d) 3^n

iv. What is the postfix expression for the infix expression $a\hat{}b\hat{}c$ ($\hat{}$ is right associative)?

a) $abc\hat{}\hat{}$ b) $ab\hat{}c\hat{}$ c) $\hat{}\hat{}abc$ d) $\hat{}ab\hat{}c$

v. What does the following function do for a given Linked List with first node as head?

```
void fun1(struct node* head){
    if(head == NULL)
      return;
    fun1(head->next);
    printf("%d ", head->data);
}
```

a) Prints all nodes b) Prints all nodes in reverse order
c) Prints alternate nodes of Linked List d) Prints alternate nodes in reverse order

vi. The Θ notation for function $f(n) = 2n^3 + n - 1$ is?

a) n b) n^2 c) n^3 d) n^4

Group B (Attempt any four questions from the following.)

2. (a) Write an algorithm/function to insert an element into an array at a given position. [2]
   (b) Write an algorithm/function to delete an element from a given position in an array. [2]
   (c) Average case time complexity for insertion algorithm. [1]

3. (a) Write the postfix evaluation algorithm. [3]
   (b) Evaluate the following postfix expression (show each step) 6 2 3 + − 3 8 2 / + *. [2]

4. (a) Write the functions for *push()* and *pop()* operations of a stack using an array in C. [3]
   (b) What are the *isFull()* and *isEmpty()* conditions for a queue using array? [2]

5. In a circular queue implemented using an array of size 7, the initial values of both front and rear are set to 0. Determine the final values of front and rear after performing the following sequence of operations: five insertions, one peek operation, two delete operations, six insertions, and three delete operations. [5]

6. (a) Write a function to insert a node at the end of the singly linked list. [2]
   (b) Write a function to search an item of the singly linked list. [3]

7. Let the size of the elements stored in an 8×3 matrix be 4 bytes each. If the base address of the matrix is 3500, then find the address of $A[5, 2]$ for both row major and column major cases in C language. [5]

ACADEMY OF TECHNOLOGY

Internal Examination, ODD Sem, 2025

Paper name: Computer Organization | **Paper Code: PCC-CS-302**
Discipline: CSE | **Semester: 3rd**
Time: 1 Hour | **Full Marks: 25**

Group A

1. Attempt any five questions from the following. **[1 × 5 = 5]**

i) What is a compiler?

ii) State the different methods of representing negative decimal numbers in binary form?

iii) What is the full form of PC in the context of computer organization?

iv) Represent the decimal number $(-11)_{10}$ in binary form using 2's complemented format.

v) The binary number $(1101)_2$ is in signed magnitude form. Calculate the decimal value of the given binary number.

vi) The Booth's algorithm is used for the multiplication between signed numbers:
a. True b. False

vii) Adding $(01101101)_2$ to $(10100010)_2$ in 8 bit 2's complement binary will cause an overflow:
a. True b. False

Group B (Attempt any four questions from the following)

2. Calculate multiplication between $(+3)_{10}$ and $(-7)_{10}$ using Booth's multiplication method. [5]

3. Draw flow chart and discuss Non-Restoring division method for unsigned numbers. [5]

4. Explain implied and memory direct addressing mode of a computer system. [5]

5. Represent the decimal number -1.75 in IEEE754 single precision format. [5]

6. Design a composite adder/subtractor circuit and explain its operation. [3+2]

7. Evaluate the arithmetic expression X = (A+B) *C in zero address and three address instruction formats. [5]

ACADEMY OF TECHNOLOGY

Internal Examination, Odd Semester, 2025

Paper name: Mathematics III | Paper Code: BSC 301

Discipline: CSE | Semester: 3rd

Time: 1 Hour | Full Marks: 25

Group A

1. Attempt any five questions from the following. [1×5 = 5]

i) The degree of the differential equation $(1+\frac{d^2y}{dx^2})^{\frac{3}{2}} = k\frac{d^2y}{dx^2}$ is

(a) 1 (b) $\frac{3}{2}$ (c) 2 (d) 3

ii) A solution of the differential equation $xdy + ydx = 0$ is

(a) $x = 3y$ (b) $x + y = 3$ (c) $x = y$ (d) $xy = 3$

iii) The number of arbitrary constants involved in the general solution of the differential equation

$2\frac{d^4y}{dx^4} + 3\frac{d^3y}{dx^3} - 8\frac{d^2y}{dx^2} + 9\frac{dy}{dx} + 2y + 6 = 0$ is

(a) 4 (b) 3 (c) 2 (d) 1

iv) The singular solution of $y = px - \frac{p^2}{4}$ is

(a) $y = x^2$ (b) $y = x - \frac{1}{4}$ (c) $y = 0$ (d) $y = 2x - 1$

v) The P.I. of the equation $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x$ is

(a) x (b) $\frac{x}{2}$ (c) $-\frac{x}{2}$ (d) $-x$

vi) Find the value of $\frac{1}{(D-2)(D-3)}e^{2x}$.

vii) Find the particular solution of

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$. When $x = 0, y = 3$ and $\frac{dy}{dx} = 0$.

Group B

(Attempt any four questions from the following.) [5×4 = 20]

2. Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$.

3. Find the general and singular solution for the equation $= px + p - p^2$.

4. Solve $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = x\,logx$.

5. Solve $(D^2 + 2D + 2)y = xe^x$.

6. Solve $\frac{d^2y}{dx^2} + a^2y = \sec ax$.

7. Solve $(D^2 - 2D + 5)y = 10sinx$.